

# Memory at your fingertips: how viscoelasticity affects tactile neuron signaling

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## eLife Assessment

The **fundamental** findings reported here provide insight into how the viscoelasticity of the fingertip skin influences the activity of mechanoreceptive afferents and thus the neural coding of force in humans. The basic principle studied was whether and to what extent the previous applied force directions impact the firing of FA-1, SA-1 and SA-2 neurons during the current applied force directions. The data and analyses are **compelling** and will be helpful for modeling the neural representations of force in the context of object grasping and manipulation.

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## Abstract

Human skin and its underlying tissues constitute a viscoelastic medium, implying that any deformation depends not only on the currently applied force, but also the recent loading history. The extent to which this physical memory influences the signaling of first-order tactile neurons during natural hand use is not well understood. Here, we examined the effect of past loading on the responses of fast-adapting (FA-1) and slowly-adapting (SA-1 and SA-2) first-order tactile neurons innervating the human fingertip to loadings applied in different directions representative of object manipulation tasks. We found that variation in the preceding loading affected neurons' overall signaling of force direction. Some neurons kept signaling the current direction, while others signaled both the current and preceding direction, or even primarily the preceding direction. In addition, ongoing impulse activity in SA-2 neurons between loadings signaled information related to the fingertip's viscoelastic deformation state. We conclude that tactile neurons at the population level signal continuous information about the fingertip's viscoelastic deformation state, which is shaped by both its recent history and current loading. Such information might be sufficient for the brain to correctly interpret current force loading and help in computing accurate motor commands for interactions with objects in manipulation and haptic tasks.

## Introduction

To enable successful skilled object manipulation and haptic object exploration, the brain must have access to information related to the forces acting on the fingertips (Johansson and Westling, 1984 [↗](#); Westling and Johansson, 1984 [↗](#); Robles-De-La-Torre and Hayward, 2001 [↗](#)). Experimental evidence indicates that populations of first-order tactile neurons with end-organs distributed throughout the fingertip skin provide sensory information about the distribution, magnitude, and direction of such fingertip forces (see Johansson and Flanagan, 2008 [↗](#), 2009 [↗](#), for reviews). However, these neurons do not signal contact forces per se, but local tissue deformations at the site of their receptor organs. This means that the relationship between fingertip forces and neuronal signaling can be very complex, since the deformation patterns resulting from a force applied to the fingertip depends on its geometry and its non-linear, viscoelastic and anisotropic material properties (Serina et al., 1997 [↗](#); Pawluk and Howe, 1999 [↗](#); Nakazawa et al., 2000 [↗](#); Jindrich et al., 2003 [↗](#); Pataky et al., 2005 [↗](#); Wang and Hayward, 2007 [↗](#)). Concerning the viscoelastic properties in particular, the recent loading history of the fingertip might influence a neuron's response to a given contact force. That is, since tissue viscoelasticity causes deformation changes to lag force changes, residual deformations from previous loadings will affect how the fingertip reacts mechanically under a given loading, and thus affect the deformation changes to which neurons' receptor organs are exposed.

Despite indications in early animal studies that stimulation history via tissue viscoelasticity may indeed affect the responsiveness of first-order tactile neurons (Lindblom, 1965 [↗](#); Werner and Mountcastle, 1965 [↗](#); Beitel et al., 1977 [↗](#); Pubols, 1982 [↗](#)), the issue has subsequently received little attention in studies of tactile mechanisms. Because the viscoelasticity of human fingertips exhibits time constants of up to several seconds (D'Angelo et al., 2016 [↗](#); Kumar et al., 2015 [↗](#); Pawluk and Howe, 1999 [↗](#); Wu et al., 2003 [↗](#)), we hypothesized that loading history would interfere with the signaling of first-order tactile neurons to rapidly fluctuating fingertip forces on a similar time scale to those experienced naturally (Kunesch et al., 1989 [↗](#); Callier et al., 2015 [↗](#); Morley et al., 1983 [↗](#)).

Here, we tested the effect of loading history on neural information transmission in human first-order tactile neurons about the direction of fingertip forces during repetitive loadings that mimic those occurring during natural object manipulation. We examined information conveyed in the three types of neurons: fast-adapting type I (FA-1), slowly adapting type I (SA-1), and slowly adapting type II (SA-2) (Birznieks et al., 2001 [↗](#); Johansson and Birznieks, 2004 [↗](#); Birznieks et al., 2009 [↗](#); Saal et al., 2009 [↗](#)). These neuron types most likely supply Meissner corpuscles, Merkel cell neurite complexes, and Ruffini-like end-organs, respectively. For neurons of all three types, we show that variations in loading history not only affect fingertip deformation, but also reduce information about the force direction in the prevailing loading. Although most neurons of each type continued to preferentially signal information about the current force direction, a minority signaled more information about the preceding force direction than the current one. For those SA-2 neurons that exhibit ongoing activity without external stimulation (Birznieks et al., 2009 [↗](#); Johansson, 1978 [↗](#); Knibestöl, 1975 [↗](#); Chambers et al., 1972 [↗](#)), we found that they could signal information about the viscoelastic state of the fingertip even when unloaded.

## Results

We recorded action potentials in the median nerve of individual low-threshold A-beta myelinated first-order human tactile neurons innervating the glabrous skin of the fingertip (Vallbo and Hagbarth, 1968 [↗](#)). Sixty of the neurons were fast adapting type I (FA-1), 73 were slowly adapting

type I (SA-1) and 41 were slowly adapting type II (SA-2) neurons (Vallbo and Johansson, 1984 [↗](#)). The receptive fields of the neurons within each class were widely distributed over the glabrous skin of the distal phalanx (**Figure 1A** [↗](#)). The fourth type of tactile neurons in the human glabrous skin, fast-adapting type II neurons (FA-2) supplied by Pacinian corpuscles, were not considered because our stimuli did not contain mechanical events with frequency components high enough to reliably excite them. Fingertip forces were applied in 5 different directions with a flat surface (referred to as the contactor), which was always oriented parallel to the skin surface at the primary site of object contact in dexterous tasks, i.e., in the middle of the flat portion of the volar surface of the fingertip. Because forces were applied to a standardized site, the neurons could vary widely in their responsiveness depending on where in the mechanically complex fingertip their transduction sites were located. Forces were applied normal (N) to the skin and at 20 degrees to the normal direction in the radial (R), distal (D), ulnar (U), or proximal (P) direction, respectively (**Figure 1B** [↗](#)). All force stimuli consisted of a force protraction phase (125 ms), a plateau phase at 4 N normal force (250 ms), and a force retraction phase (125 ms); inter-stimulus periods were 250 ms.

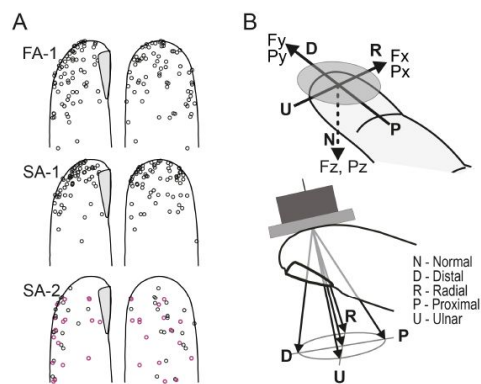
Each neuron was subject to two different sequences of force stimuli, first the ‘regular sequence’ and then the ‘irregular sequence’. As stimuli were force controlled, the contactor’s movement and position producing identical reactive force may differ depending on stimulation history, due to viscoelastic properties of the fingertip tissue. In the regular sequence, the five force directions were repeatedly presented in a fixed relative order (R, D, U, P, N), such that each loading in each direction received the same immediate stimulation history (**Figure 2A** [↗](#)). Differences in the path of the contactor between different force directions were clearly visible in the tangential plane, where the changes in the force stimulation took place between trials (**Figure 2B** [↗](#)). In the regular sequence, the contactor path was practically identical across the five test trials in each direction (**Figure 2B** [↗](#)), indicating that the fingertip deformed similarly during the repetitions. However, the contactor path deviated from the direction of the force due to anisotropic mechanical properties of the fingertip and, importantly, it differed between force protraction and retraction due to the viscoelasticity of the fingertip (**Figure 2B** [↗](#)). Likewise, the viscoelastic properties were reflected as a pronounced hysteresis between the force and the contactor position, creep during the force plateaus, and creep recovery during the interstimulus periods (**Figure 2A** [↗](#)).

In the irregular sequence, also including five trials in each force direction, the stimulation history for trials in each direction varied systematically, such that trials in each stimulation direction were preceded by loading in each of the five directions once (**Figure 2C** [↗](#)). As in the regular sequence, the contactor moved broadly in the direction of the force regardless of the previous loading direction, but its precise path differed markedly depending on the previous loading direction (**Figure 2D** [↗](#)). This variability was most evident during the protraction phase (**Figure 2E** [↗](#)) but could be discerned throughout the trial (**Figure 2D** [↗](#)).

Hence, deduced from the contactor’s behavior, these observations indicated that variations in the immediate loading history caused a greater intertrial variation in the fingertip deformation than an unchanging loading history. This is illustrated in **Figure 3A** [↗](#), which shows the contactor path for both regular and irregular sequences from a single participant for five test trials in the distal direction as well as the corresponding preceding trials.

## Effects of loading history on neural responses

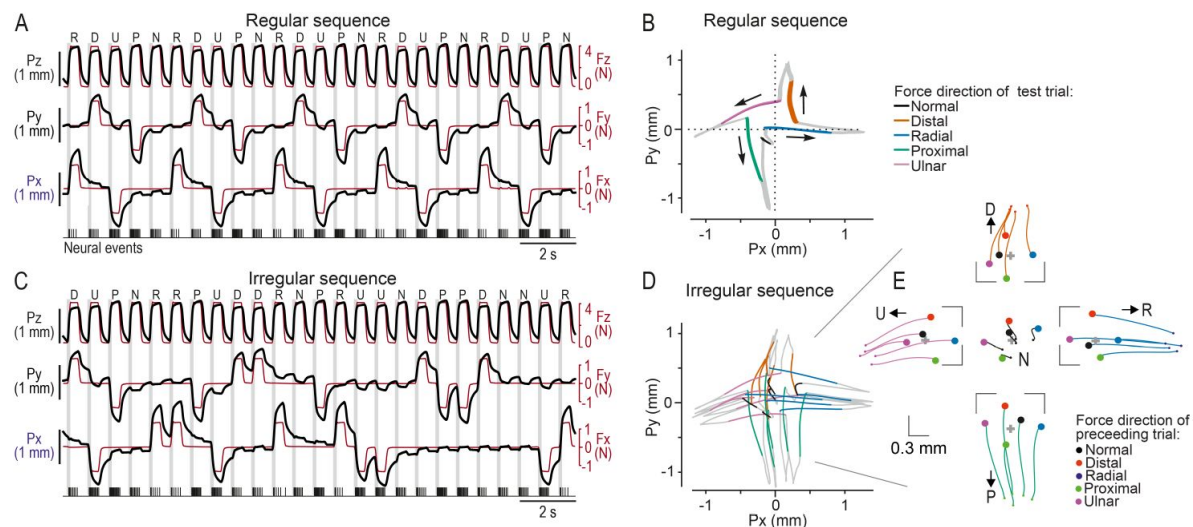
We asked whether the greater intertrial variability in fingertip deformation during the irregular stimulation sequence would be reflected in the responses of tactile neurons. Indeed, we observed greater intertrial variability in the firing rate profiles of the neurons in the irregular compared to the regular sequence during test trials (see examples in **Figure 3B–D** [↗](#)). This increased variability was most evident during the force protraction phase where most neurons exhibited the most intense responses.



**Figure 1.**

### Experimental setup.

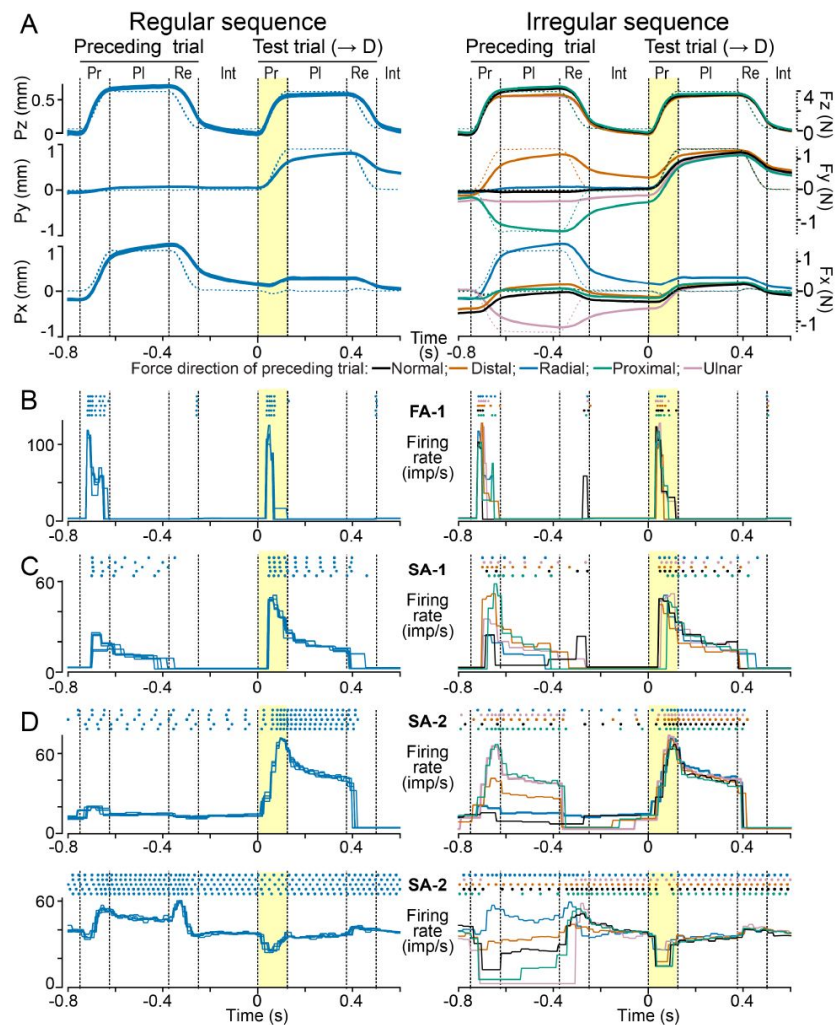
**A.** Receptive field center locations shown on a standardized fingertip for all first-order tactile neurons included in the study, categorized by neuron type. Purple symbols denote spontaneously active SA-2 neurons exhibiting ongoing activity without external stimulation. **B.** The flat stimulus surface was centered at the standard site of stimulation and oriented such that its tangential plane was parallel to the flat portion of skin on the fingertip. The stimulus maintained contact with the skin at a force of 0.2 N in intertrial periods. Force stimuli were superimposed on this background contact force and were delivered in the normal direction (N), and at an angle of 20 degrees to the normal with tangential components in the distal (D), radial (R), proximal (P) and ulnar (U) directions, as indicated by the five arrows in the lower panel.



**Figure 2.**

### Stimulation sequence exposes fingertip viscoelasticity.

**A.** Trial order for the entire regular sequence, which repeats fingertip loadings in five different force directions in a fixed order, implying that loadings in each direction received the same stimulation history. Force (red lines) and contactor position (black lines) are shown along the normal (z), distal/proximal (y), and ulnar-radial (x) axes, while recording action potentials from a single exemplary SA-1 neuron (bottom trace). **B.** Average contactor position in the tangential plane for all trials in the regular sequence across all recorded neurons (and thus fingertips). The colored segments of the curves indicate the protraction phase for each of the five force directions, while other phases of the fingertip loading (plateau, retraction) and the inter-stimulus period are shown in grey. Dashed lines show the directions in which the tangential force components were applied. **C.** Trial order for the entire irregular sequence, where force directions are varied such that trials in each stimulation direction were preceded by loading in each of the five directions once. Same neuron (and fingertip) as in A. **D.** Average contactor position in the tangential plane for the irregular sequence. Same format as in B. **E.** Contactor position in the tangential plane at the start of (filled circles) and during the force protraction phase (colored lines) per force direction, referenced to the fingertip position at rest (gray marker). Same trials as in D, but different force directions are shown in separate panels for better visibility.



**Figure 3.**

### Influence of preceding loading direction on fingertip deformation and neural responses.

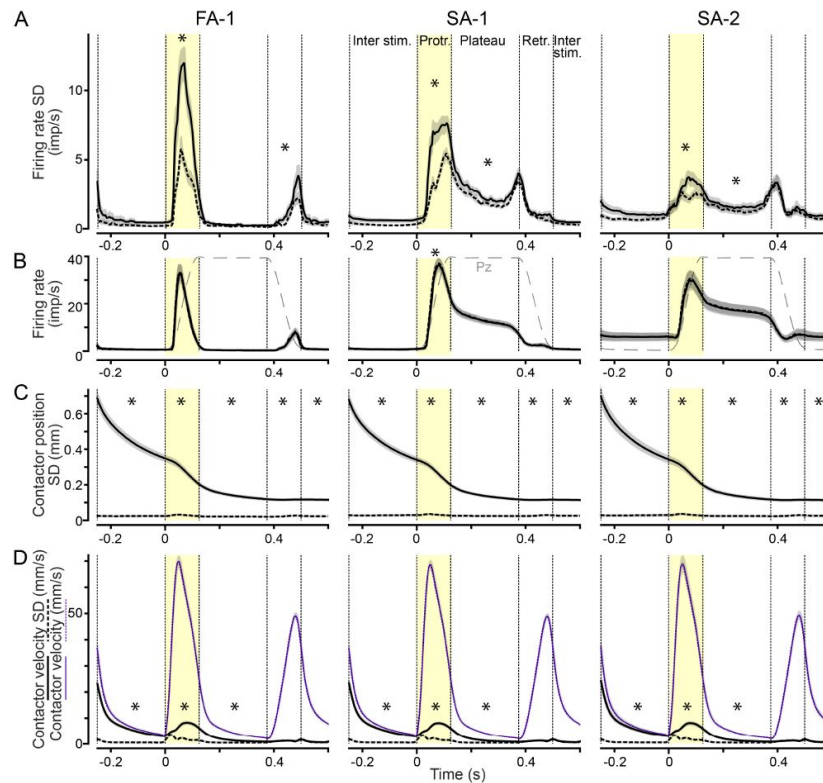
**A.** Contact position along the x, y, and z axes (see [Figure 1B](#)) as a function of time superimposed for all five trials with loading in the distal direction ('test trial') as well as the respective previous loading ('preceding trial') in the regular (left column) and the irregular (right column) sequence. Vertical dashed lines mark transitions between loading phases (Pr: protraction, Pl: plateau, Re: retraction phase, Int: intertrial period). The yellow shaded area indicates the protraction phase of the test trials. Each trace is colored according to the force direction of the previous loading. Data were recorded from a neuron whose response is shown in **B**. **B.** Dots (top) represent action potentials recorded from an FA-1 neuron for each trial, whose contactor movements are shown in **A**. The superimposed traces below represent the corresponding firing rate profiles, defined as the reciprocal of the interval between subsequent action potentials. Color coding as in **A**. **C,D.** Exemplary responses of one SA-1 (**C**) and two SA-2 neurons (**D**) to force loadings corresponding to those in **A**. All neurons show higher variability in their firing rate profiles during test trials in the irregular compared to the regular sequence.

Increased variability was also observed in instances where the dynamic response to force stimulation involved a decrease in the firing rate (lower panels of **Figure 3D** [↗](#)). This phenomenon was observed in SA-2 neurons that maintained an ongoing discharge during intertrial periods (cf. **Figure 1A** [↗](#)). In these cases, the response to a force stimulus constituted a modulation of the firing rate around the background discharge, signifying that a force stimulus could either decrease or increase the firing rate depending on the prevailing stimulus direction. For SA-1 and SA-2 neurons, which typically generated nerve impulses also during the force plateau, and for FA-1 neurons, which often responded during force retraction, we noted a tendency for greater variability in the intensity of these responses as well. Finally, the variation in previous loading direction during the irregular sequence could also modify impulse activity in SA-2 neurons generated during interstimulus periods, the implications of which will be further addressed below.

We quantified the effect of the stimulation history on a neuron's response by first calculating for each stimulation sequence the time-varying standard deviation of the instantaneous firing rate during the test trials in each of the five force directions. Averaged across neurons and loading directions, firing rate variability during the force protraction phase was just over twofold higher for FA-1s in the irregular sequence compared to the regular sequence, almost twofold higher for SA-1 and about 70% higher for SA-2 neurons (**Figure 4A** [↗](#)). In addition to the protraction phase, variability was elevated in the irregular sequence for both SA types during the plateau phase and for FA-1 neurons during the retraction phase (all  $p_{corrected} \leq 0.001$ , paired Wilcoxon signed rank tests). FA-1 neurons did not respond during the plateau phase (see **Figure 4B** [↗](#)), while both SA types responded only weakly during the retraction phase with no apparent difference between the sequences ( $p \geq 0.3$ ). Thus, in all phases where neurons responded reliably, their response variability in test trials increased when the force direction in the preceding trial varied. Notably, neurons' overall firing rates did not differ between the regular and irregular sequence when averaged over test trials in all force directions (**Figure 4B** [↗](#)) other than for SA-1s during the protraction phase ( $p_{corrected} = 0.002$ ), where the difference in firing rate was less than 1 imp/s. This suggested that the greater variability was linked to the stimulation history and not a change in the neurons' overall responsiveness.

We also analyzed the effect of stimulation sequence on the fingertip deformation, again by using the contactor behavior as a proxy. For each neuron examined and stimulation sequence, we calculated the time-varying standard deviation in the tangential plane of the contactor position and of its velocity across the five test stimuli in each of the five force directions. As results were similar across the different test directions, we then averaged these data to arrive at a single variability measure per neuron. Importantly, the variability in contactor position was markedly higher in the irregular than in the regular sequence (**Figure 4C** [↗](#), solid vs. dashed lines). The variability in the irregular sequence decreased over time and did so especially rapidly during the force protraction phase. Averaged across all neurons, at the beginning of the force protraction the variability in contactor position was about 20 times greater in the irregular than in the regular sequence.

Variability was 7 times greater during the plateau phase and still about 4 times so at the end of the retraction phase. During all phases, including the interstimulus period before and after the test trial, the variability in the irregular sequence was significantly higher than in the regular sequence ( $p_{corrected} < 0.001$ , paired Wilcoxon signed rank tests). The effect of stimulation sequence present even at the end of the force retraction phase indicates that the viscoelastic memory trace of the previous loading direction lingered to some extent even beyond the test trials. Notably, the time-varying variation in contactor position was virtually identical for data pertaining to each of the three neuron types (**Figure 4C** [↗](#)), demonstrating that the differences in neural response behavior could not be explained by variability in the skin responses across the different experimental runs.



**Figure 4.**

**Increased variability in neural responses and fingertip deformations in the irregular sequence.**

**A.** Standard deviation of neuronal instantaneous firing rates as a function of time during test trials in each loading direction in the regular (dashed lines) and irregular (solid lines) sequences. Data averaged across neurons of each type and all loading directions. Shaded areas indicate SEM. Vertical dashed lines mark transitions between loading phases as in [Figure 3A](#) and stars indicate phases where there was a significant difference between regular and irregular sequences at  $p < 0.05$ . **B.** Average instantaneous firing rates for the regular and irregular sequence as a function of time. Dashed black lines indicate the force profile of the fingertip loading. Note that the average firing rates are almost identical in the regular and irregular sequence. **C.** Standard deviation of tangential (2D) contactor position for the regular and irregular sequences. **D.** Standard deviation of contactor velocities (black lines) and average contactor velocity (purple lines) for the regular and irregular sequences.

Regarding the effect of viscoelasticity variability on contactor velocity, we found that the variability in contactor velocity was significantly greater in the irregular than in the regular sequence during the initial interstimulus period, the force protraction phase, and the plateau phase (**Figure 4D** [↗](#), black curves;  $p_{corrected} < 0.001$ ), but not during the retraction phase or the subsequent interstimulus period ( $p > 0.4$ ). However, even during the irregular sequence, the variability in contactor speed appeared rather modest compared to the absolute contactor speed (**Figure 4D** [↗](#), cf. black and gray curves), which likely primarily drove the dynamic neural response during the force protraction for all classes (cf. **Figure 4B** [↗](#)).

Taken together, the variable loading history in the irregular sequence affected the neurons' firing rates and most so during their dynamic responses elicited by the force protraction. Further, the effects on the neurons' responses reasonably matched the influence of the variation in previous force direction on the state of the fingertip deformation and its change during the test trials, which was also most pronounced during the force protraction phase.

## Predicting neural responses from contactor movements

The similarity in the history-dependent variation in neural firing and fingertip deformation at a given force stimulus suggests that neuronal firing is determined by how the fingertip deforms rather than the applied force itself. However, this similarity does not clarify the relationship between fingertip deformation dynamics and neural signaling. To investigate further, we fit cross-validated multiple linear regression models to evaluate how well distinct aspects of contactor movement could predict the time-varying firing rates of individual neurons during the protraction phases of the irregular sequence. The models used predictors based on (1) the three-dimensional position of the contactor, (2) its three-dimensional velocity, (3) a combination of position and velocity signals, and, finally, (4) position and velocity signals along with all possible two-way interactions between them, capturing potentially complex relationship between fingertip deformations and neural signaling.

Comparing the variance explained ( $R^2$ ) by each regression model for each neuron type revealed clear differences between the models (**Figure 5A** [↗](#)). A two-way mixed design ANOVA, with regression model as within-group effects and neuron type as a between-group effect revealed a main effect of model on variance explained ( $F(3, 462) = 815.5$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.84$ ). Model prediction accuracy overall increased with the number of predictors, with the two-way interaction model outperforming all others ( $p < 0.001$  for all comparisons, Tukey's HSD). Additionally, a significant main effect of neuron type ( $F(2, 154) = 29.8$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.28$ ) and a significant interaction between regression model and neuron type were observed ( $F(6, 462) = 50.8$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.40$ ).

For neuron type, model predictions were most accurate for SA-2 neurons, followed by SA-1 neurons, with FA-1 neurons showing the lowest accuracy ( $p < 0.003$  for all comparisons, Tukey's HSD). The interaction between model and neuron type revealed distinct patterns. For SA-1 and SA-2 neurons, position-only and velocity-only models had similar prediction accuracy ( $p \geq 0.996$ , Tukey's HSD) with no significant differences between these neuron types ( $p \geq 0.552$ , Tukey's HSD). FA-1 neurons performed poorly with the position-only model but showed higher accuracy with the velocity-only model ( $p < 0.001$ , Tukey's HSD) and better than SA-1 neurons ( $p = 0.006$ , Tukey's HSD). Models combining position and velocity predictors (without interactions) surpassed both position-only and velocity-only models for SA-1 and SA-2 neurons ( $p < 0.001$ , Tukey's HSD). Overall, the differences between neuron types broadly match their tuning to static and dynamic stimulus properties.

The two-way interaction model, accounting for most variance in neural responses, produced mean  $R^2$  values of 0.75 for FA-1, 0.88 for SA-1, and 0.91 for SA-2 neurons (**Figure 5A** [↗](#)). To evaluate the contribution of the different predictors, we ranked them using the permutation feature importance method, focusing on the six most important ones. Regression analyses using only these variables explained almost all of the variance explained by the full model, with a median  $R^2$

Figure 5.

Neurons' sensitivity to different aspects of contactor movement.

**A.** Variance explained ( $R^2$ ) when predicting time-varying firing rates using four regression models: (1) 3D contactor positions ('pos'), (2) 3D velocities ('vel'), (3) a combination of positions and velocities ('pos + vel'), and (4) a model incorporating all two-way interactions ('2-way int') for FA-1, SA-1, and SA-2 neurons. The interaction model consistently outperformed the others across all neuron types. **B.** Relative frequency of neurons, categorized by type, for which a given predictor ranked among the six most important in the two-way interaction model. For a detailed breakdown of predictor importance by rank, see Figure 5—Figure Supplement 1 [🔗](#).

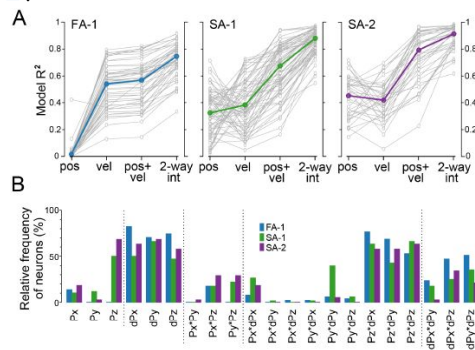
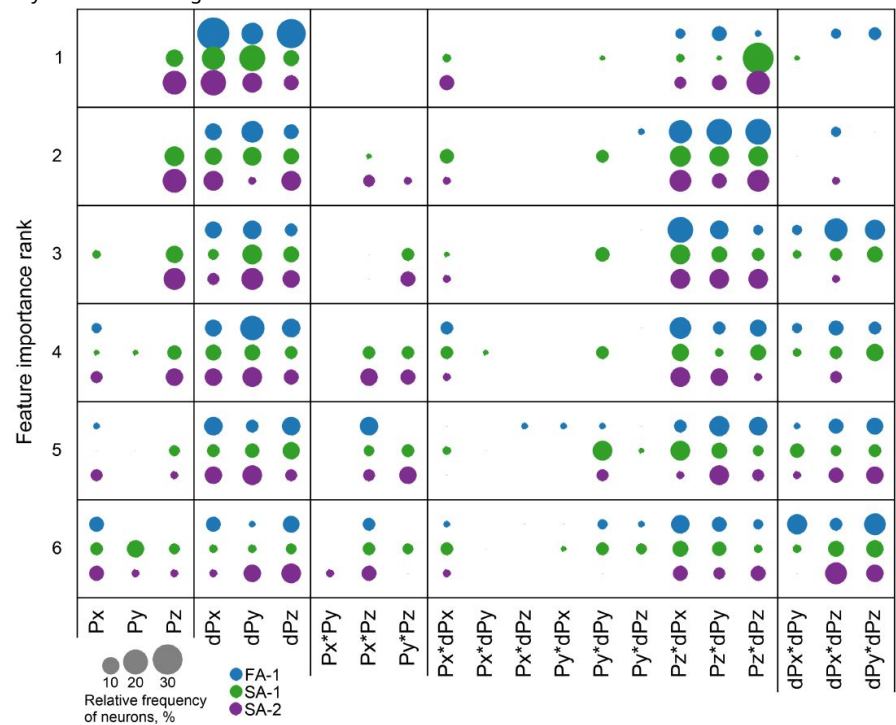


Figure 5—Figure supplement 1.

Importance of individual predictors in the two-way interaction model.

Importance rank (rows 1–6) of each predictor (columns) for each neuron type (color-coded), with circle areas proportional to the relative frequency of neurons assigned to each rank.



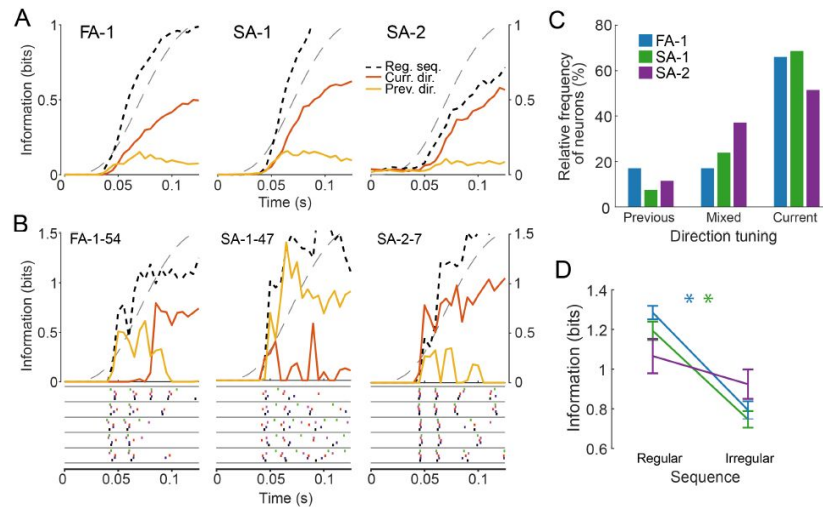
reduction of just 0.055 across all neurons. Across all neuron types, at least half included all three velocity components (dPx, dPy, dPz) among the top six, with FA-1 neurons showing the highest prevalence (**Figure 5B**). Interactions between normal position (Pz) and each velocity component were also frequently observed, while interactions involving tangential position and velocity components were less common. Interactions among velocity components were relatively well represented, followed by interactions limited to position components. Position signals were generally less represented, except for normal position (Pz) in slowly adapting neurons, where it appeared in 50% of SA-1 and 68% of SA-2 neurons. Despite these broad trends, important predictors varied widely across ranks even within a given neuron class (see **Figure 5—Figure Supplement 1**), and even the most frequent variables appeared in only a subset of cases, suggesting broad variability in sensitivity across neurons.

## Information transmission about past and present loading

To assess how the increased firing rate variability observed in the irregular sequence as well as the variation in sensitivity to different aspects of contactor movement affected neural information transmission, we calculated a lower bound on the mutual information transmitted about both the current and the preceding force direction based on the neural spiking responses of individual neurons (see Methods). We focused on protraction phase only, during which firing rates and their variability were highest.

Averaged across all neurons of each type, information about the current force direction tended to accumulate throughout the force protraction phase, as shown previously for FA-1 and SA-1 neurons (Saal et al., 2009). However, for all types, the rate of information increase was considerably lower and cumulative information tended to plateau at a much lower value in the irregular than in the regular sequence (compare black dashed with orange lines in **Figure 6A**). At the end of the protraction phase, both FA-1 and SA-1 neurons signaled on average only 50% of the information in the irregular compared to the regular sequence ( $p < 0.001$  for both types, paired Wilcoxon signed rank tests). For the SA-2 neurons the corresponding information transfer was reduced by only 20% ( $p = 0.02$ ). However these neurons signaled far less information about force direction to begin with. As a result, average information transmission in the three classes of neurons ended up comparable in the presence of viscoelastic effects.

We also assessed information about the preceding force direction contained in the neural responses during the protraction phase in the irregular sequence. We reasoned that if the preceding loading systematically affected contactor position in the subsequent trial, then neurons might carry information about past stimulation in their responses. Consistent with this idea, such information was present, albeit at a relatively low level. Averaged across all neurons of each type, information about the preceding force direction increased for about 60–70 ms into the protraction phase after which it appeared to plateau or decrease (yellow traces in **Figure 6A**;  $p < 0.001$  for each type, Wilcoxon one-sample signed rank tests based on information values halfway through the protraction phase compared against zero information). Irrespective of neuron type, signaling of the preceding as well as the current force direction could vary substantially between neurons, with some carrying information mostly about the current direction, and others about the preceding one (see example information traces for individual neurons in **Figure 6B**). We quantified the diversity amongst neurons in this respect based on whether they primarily conveyed information about the present force direction, the preceding force direction, or a mix of both (including 53 out of 67 FA-1, 67 out of 73 SA-1 and 35 out of 41 SA-2; see Methods). Most neurons primarily signaled information about the current force direction (46 SA-1, 35 FA-1 and 18 SA-2 neurons, see **Figure 6C**). Fewer showed mixed tuning and those that did signaled preceding force direction early during the protraction phase and then switched to information about the current loading direction (16 SA-1, 9 FA-1 and 13 SA-2). Finally, some neurons (9 FA-1, 4 SA-1 and 4 SA-2) primarily signaled information about preceding force direction. We found no significant differences in the relative frequency of FA-1, SA-1 and SA-2 neurons that responded to the current or previous stimulation, or both ( $\chi^2(3, 155) = 6.95$ ,  $p = 0.14$ ). Notably, information transmission in



**Figure 6.**

### Information about current and previous force direction during the protraction phase.

**A.** Average mutual information about force direction for FA-1 (left), SA-1 (middle), and SA-2 (right) neurons as a function of time during the protraction phase in the irregular sequence. Information is shown for the current trial (solid orange line) or the preceding trial (solid yellow line) and is compared to the regular sequence (dotted black line). Grey dashed lines denote the stimulus force profile. **B.** Mutual information curves (top) and spike trains (bottom) for three individual neurons with different response behaviors. Information curves as in A. Spike trains are split by current force direction and colored by previous force direction. Examples are of a neuron with mixed tuning (left), a neuron that signals information about the previous stimulus (middle), and a neuron that signals information about the current one (right). **C.** Proportion of FA-1 (blue), SA-1 (green), and SA-2 (purple) neurons showing different response behaviors during the protraction phase. Most neurons signaled the current force direction, but around 35% either signaled the previous force direction or showed mixed response behavior. **D.** Information transmitted about force direction for neurons tuned to the current force direction for the regular and the irregular sequence. Information decreases considerably for the irregular sequence, even in neurons responding strongly to the current direction.

both FA-1 and SA-1 neurons decreased between the regular and irregular sequence even when only considering those neurons that predominantly signaled information about the current force direction (**Figure 6D**), confirming that information transmission about the ongoing stimulus was still affected by the fingertip's viscoelastic memory ( $p < 0.001$  for both types,  $p = 0.18$  for SA-2, paired Wilcoxon signed rank tests). We found no straightforward relationship between a neuron's sensitivity to current and previous stimulation and its termination site in fingertip skin. Specifically, there was no statistically significant effect of the distance between a neuron's receptive field center and the primary contact site of the stimulus surface on whether neurons signaled current, prior, or mixed information for SA-1 (Kruskal-Wallis test,  $H(2) = 3.86$ ,  $p = 0.15$ ) or SA-2 neurons ( $H(2) = 0.75$ ,  $p = 0.69$ ). However, a significant difference emerged for FA-1 neurons ( $H(2) = 8.66$ ,  $p = 0.01$ ), indicating that neurons terminating closer to the stimulation site on the flat part of the fingertip were more likely to signal past or mixed information.

Overall, our findings suggest that past loading reduces the information conveyed by these neurons about the direction of prevailing fingertip forces. Nevertheless, neurons of all three types retain information about the previous force direction, albeit with substantial variability among individual neurons. This variability may be partially attributed to the specific locations on the fingertip where neurons terminate, determining the local strain patterns they experience. Additionally, differences in the specific aspects of local strain to which individual neurons respond could further contribute to this heterogeneity.

## Neural responses in the interstimulus period

SA-2 neurons can exhibit ongoing activity with-out external stimulation and sense tension states in collagenous fiber strands in dermal and subdermal tissues (Knibestöl, 1975; Birznieks et al., 2009; Johansson, 1978; Chambers et al., 1972). Since the deformation of the fingertip by the force stimuli was mostly absorbed by such tissues, we hypothesized that SA-2 neurons active during the interstimulus periods might convey ongoing information about the viscoelastic state of the fingertip during the recovery from the recurrently applied loadings. A subset of SA-2 neurons (20 out of 41) exhibited such activity (see **Figure 7A** for three examples). Calculation of mutual information indicated that some individual neurons were highly informative about the preceding stimulus direction, but the time of maximal information transmission could occur at different points (see examples in **Figure 7B**). On average, SA-2 neurons provided low but continuous information about the preceding force direction throughout the interstimulus period, which was highest at the start of the interstimulus period and tended to decrease slightly over time (**Figure 7C**, yellow trace). This decrease was likely driven by the gradual relaxation of the fingertip. We next asked whether SA-2 neural activity at the population level could track the deformation state of the fingertip during the interstimulus period. We calculated a low-dimensional representation of the SA-2 population activity to compare with the fingertip deformation at three different time points: in the middle and at the end of the inter-stimulus period, and again at the end of the protraction phase. Specifically, based on activity recorded during the irregular sequence we calculated pairwise spike distances across all 25 trials (5 force directions  $\times$  5 trials) for each neuron, providing a measure for how distinct the activity of this neuron was across different trials. This process yielded a matrix, which was averaged across all active SA-2 neurons. We then used multi-dimensional scaling to place each of the 25 types of trials into a two-dimensional space. Using Procrustes analysis, finally we rotated and scaled the responses to match the recorded tangential contactor positions, which were calculated for the same 25 types of trial at the corresponding time points and averaged across fingertips (**Figure 8A**). Notably, there was a good match between the neural representation and the fingertip deformation. For both measures, different trial types were clustered according to the force direction of the preceding loading throughout the interstimulus period, but cluster separation decreased as time progressed. Then trial types diverged during the protraction phase, and at the end of the phase they were clustered according to the current force direction. To quantify the similarity of the two representations, for each we calculated two measures. The 'total variance' across all 25 trial types indicates the general level of variability (see black dashed ellipse in the top panel of **Figure 8A**). The 'direction

variance', which was calculated over trials representing the same preceding force direction and then averaged, indicates variability within clusters of trials with the same preceding direction (see dashed orange ellipse in the top panel of **Figure 8A** [↗](#)). Skin positions and SA-2 population activity displayed a similar variance pattern (**Figure 8B** [↗](#)). During the interstimulus period the total variance was much larger than the direction variance, signifying a marked clustering according to the preceding force direction, but this difference tended to decrease with time. In contrast, both variances were large and roughly equal at the end of the protraction phase, indicating little clustering based on the preceding force direction. Taken together, these results suggest that SA-2 neurons can continuously signal the mechanical state of the fingertip even in the absence of externally applied stimulation.

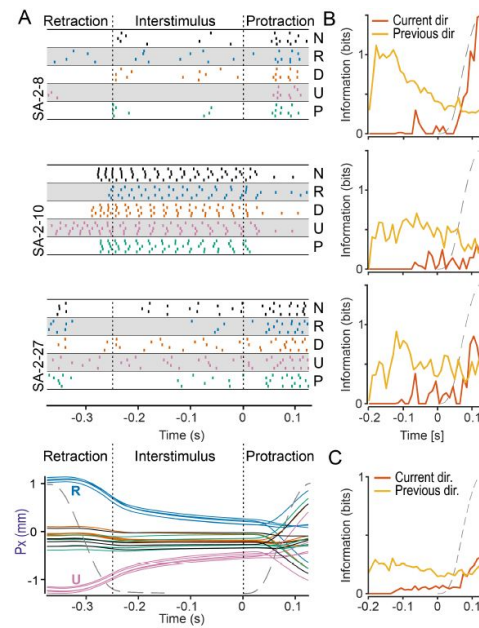
## Discussion

We found that the viscoelasticity of the fingertip affects signals in first-order tactile neurons when responding to fingertip loadings mimicking those experienced in everyday object manipulation tasks. Such tasks involve applying forces of different magnitudes and directions in rapid succession, such as during grasping and transporting objects, handicraft, cooking, cleaning, and food gathering. The neurons' signaling of force direction was significantly influenced by the direction of the preceding loading, even if it varied by only 20 degrees relative to the perpendicular direction. This effect was most prominent during the force protraction phase, which is when neurons typically signal the most information about force direction. However, there was heterogeneity in how individual neurons behaved. Some neurons primarily signaled information about the current direction, while others signaled both the current and previous direction, or even primarily the preceding direction. Our results also indicate that neurons can signal information related to the fingertip's viscoelastic deformation state even between loadings: SA-2 neurons' ongoing impulse activity between loadings was influenced by the previous loading direction. This diversity suggests that, at the population level, first-order neurons carry information about the fingertip's current viscoelastic state, which within it contains a memory of past stimulation.

### Origins of neural heterogeneity

The observed heterogeneity between neurons of the same type is expected, given that their transduction sites were distributed widely within the fingertip skin and they are sensitive to the local stresses and strains at their transduction site, rather than the global deformation of the fingertip (Birznieks et al., 2001 [↗](#); Saal et al., 2009 [↗](#)). Thus, the viscoelastic memory of the preceding loading would have modulated the pattern of strain changes in the fingertip differently depending on where their receptor organs are situated in the fingertip.

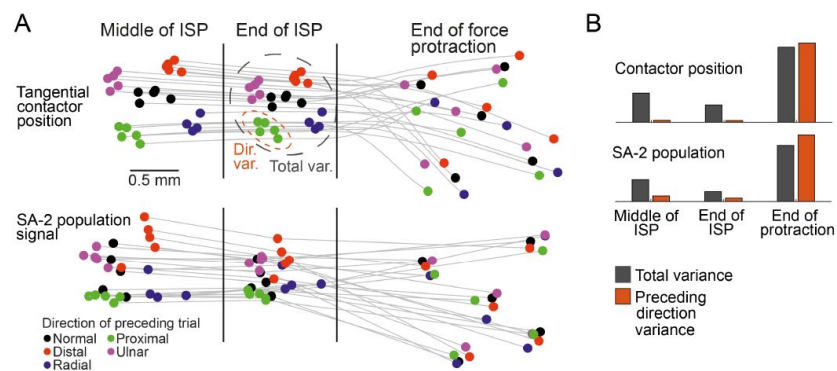
In addition to the receptor organ locations, the variation in sensitivity among neurons to fingertip deformations in response to both previous and current loadings would stem from the fingertip's geometry and its complex composite material properties. Possible inherent directional preferences of the receptor organs, attributed to their microanatomy, could also be significant. However, mechanical anisotropy, particularly within the viscoelastic subcutaneous tissue of the fingertip induced by intricately oriented collagen fiber strands forming fat columns in the pulp (Hauck et al., 2004 [↗](#)), are likely to play a crucial role. This anisotropy would shape the dynamic pattern of strain changes at neurons' receptor sites, intricately influencing a neuron's sensitivity not only to current but also to preceding loadings. Indeed, recent modeling efforts suggest that such mechanical anisotropy strongly influences the spatiotemporal distribution of stresses and strains across the fingertip (Duprez et al., 2024 [↗](#)). The resulting diversity in the sensitivities of neurons might enhance the overall information collected and relayed to the brain by the neuronal



**Figure 7.**

### SA-2 neurons signal information about previous forces in the interstimulus period.

**A.** Spike raster plots for three SA-2 neurons recorded during the interstimulus periods of the irregular sequence. Each dot represents an action potential, and the colors indicate the preceding force direction. Note that the responses differ systematically based on the previous force direction. Bottom panel: To illustrate the effect of previous force direction on the deformation state of the fingertip during the interstimulus period, the average contactor position in the ulnar-radial (U-R) direction during the corresponding irregular stimulation sequences is shown (on the same time scale as upper panels). Colors indicate the force direction in the preceding trial corresponding to the color coding in A. **B.** Average mutual information about force direction in the test trial (orange line) or the preceding trial (yellow line) during the interstimulus period and during the subsequent protraction phase for the same three neurons shown in panel A. **C.** Average mutual information across all SA-2 neurons active in the interstimulus period.



**Figure 8.**

### Continuous representation of fingertip viscoelastic state in the SA-2 population.

**A.** Colored dots indicate tangential contactor positions (top) and their representation in the SA-2 population signal (bottom) at three different times: the middle of the interstimulus period ( $-0.125$  s), the end of the interstimulus period ( $0$  s) and the end of the subsequent protraction phase ( $0.125$  s). The colors of the markers denote the force direction in the directly preceding trial. Contactor position is the two-dimensional position in the tangential plane. SA-2 representations are derived from average spike trains distances across the different trials, visualized in a two-dimensional space using multidimensional scaling and aligned with the contactor positions using Procrustes analysis (see Methods for details). **B.** Averaged total variance and within-direction variance for contactor position and SA-2 population signal representations at the same three time points as in A. As illustrated by the dashed ellipses in A, total variance denotes the two-dimensional variance across all trials, while within-direction variance denotes the variance for trials belonging to the same preceding force direction. Higher total than within-direction variance indicates that data for trials in the same preceding direction are more clustered than data for trials in all preceding directions, which is required for discrimination of preceding force direction.

population, facilitating the discrimination between tactile stimuli or mechanical states of the fingertip (see [Rongala et al., 2024](#); [Corniani et al., 2022](#); [Tummala et al., 2023](#), for more extensive explorations of this idea).

Our regression analysis, which examined how well different aspects of contactor behavior could predict the time-varying firing rates of individual neurons, highlights the complexity of neural responses to fingertip mechanics. This conclusion is supported by three key findings. First, models incorporating interactions between contactor position and velocity outperformed simpler models. Second, the importance of independent variables varied among neurons of the same type, reflecting differences in sensitivity to contactor position, velocity, and their interactions. Third, prediction accuracy varied significantly across neurons of the same type. The limited accuracy of the two-way interaction model in replicating detailed firing rate profiles of individual neurons underscores the intricate relationship between fingertip events and neural activity. Although a three-way interaction model was evaluated, it provided only marginal improvements in variance explained (data not shown).

Predictors that showed consistently high importance across neurons were position-velocity interactions, revealed by the two-way interaction model. This finding suggests that neuronal responses to fingertip deformation changes (reflected by contactor velocity) are significantly influenced by the fingertip's current deformation state (indicated by contactor position), modulated by previous fingertip loading through viscoelastic effects. Direct observations support this conclusion. First, variation in neuronal response to force protraction, depending on previous loading direction, paralleled changes in contactor position and path, whereas contactor velocity, the primary driver of the neuronal response, remained generally aligned with the force direction (cf. **Figure 2C-E**). Second, FA-1 neurons, which displayed minimal sensitivity to contactor position alone, exhibited greatly improved prediction accuracy when position and velocity variables interacted, demonstrating that the fingertip's deformation state modulates the neurons' dynamic responses.

## Differences between neuron types in sensitivity to viscoelasticity

We also observed some differences between neuron types regarding how variation in the preceding force direction affected their signaling during the current loading. The impact of stimulation history during the force protraction phase was more pronounced in FA-1 and SA-1, compared to SA-2 neurons. It is possible that the neurons' response properties accounted for this difference: type I neurons are primarily sensitive to deformation of the fingerprint ridges ([Sukumar et al., 2022](#); [Jarocka et al., 2021](#)), while SA-2 neurons primarily signal tension states in deeper dermal and subdermal tissues ([Knibestöl, 1975](#); [Birznieks et al., 2009](#); [Johansson, 1978](#); [Chambers et al., 1972](#)). The fingertip deformation changes during the loadings can be seen as twofold. First, the force applied by the contactor induced bulk deformation changes of the fingertip that depended on the force direction. As pressure increases in the pulp, the pulp tissue bulges at the end and sides of the fingertip. Simultaneously, the tangential force component amplifies the bulging in the direction of the force while stretching the skin on the opposite side. This effect is attributed to the fact that the friction between the stimulation surface and the skin was high enough to prevent it from sliding over the fingertip. Bulk deformation changes are closely linked to widespread alterations in stress and strain distribution in deeper tissues, making the SA-2 neuron population the preferred signaling source for such changes. For example, it has been demonstrated that experimentally induced compliance changes of the finger pulp using venous occlusion readily influenced activity of SA-2, but not SA-1 neurons ([Hudson et al., 2015](#)). However, concurrent with the bulk deformation changes, the fingertip skin would undergo direction dependent surface deformations changes involving planar tensile strain changes and partial slippage peripherally within the contact surface ([Delhay et al., 2014, 2016](#); [Willemet et al., 2021](#)). Changes in planar tensile strain per se might excite neurons of either type, given that even neurons terminating outside the contact area can respond to fingertip loadings ([Bisley et al., 2000](#); [Birznieks et al., 2001](#)). However, partial slippages occurring during fingertip

loadings, where some parts of the fingertip-object interface slip while others remain stuck, excite SA-1 and especially FA-1 neurons most intensely (Johansson and Westling, 1987 [↗](#); Srinivasan et al., 1990 [↗](#); Khamis et al., 2014 [↗](#); Delhay et al., 2021 [↗](#)). A substantial part of the sensitivity of FA-1 and SA-1 neurons to stimulation history could therefore be attributed to the preceding force direction affecting the location and timing of partial slips. That is, due to the viscoelastic memory of the fingertip, previous loadings in different directions could result in different patterns of planar tensile stress changes under a given loading condition, which would affect how and where the local partial slips occur. The responsiveness of type I neurons to partial slippage might also explain their apparently higher sensitivity to stimulation history compared to SA-2s. In sum, we believe that both bulk and superficial deformation changes play a role in the activation of the tactile neurons during the fingertip loadings.

Our regression analysis of contactor behavior and time-varying firing rates corroborates that the different neuron types exhibit varying sensitivities to mechanical fingertip events during fingertip loadings. The observation that predictions were most accurate for SA-2 neurons aligns with the understanding that contactor behavior during loading primarily reflects bulk deformation changes, best signaled by SA-2 neurons. The lower prediction accuracy for SA-1 and, particularly, FA-1 neurons suggests that these are influenced by additional factors that are less effectively or not represented in the behavior of the contactor, specifically, direction-dependent planar tensile strain changes in the skin and partial slippages, which likely predominantly affect FA-1 neurons (see e.g. Delhay et al., 2021 [↗](#)).

## Continuous signaling of viscoelastic state by SA-2 neurons

That SA-2 neurons signaled the viscoelastic state of the fingertip in periods between loadings is consistent with previous ideas that SA-2 neurons continuously measure stresses in collagen fiber strands that run within and between dermal and subdermal tissues (Vallbo and Johansson, 1984 [↗](#); Chambers et al., 1972 [↗](#); Birznieks et al., 2009 [↗](#)). Indeed, their relatively low dynamic sensitivity to externally applied loads and the well sustained response to maintained loadings suggest that SA-2 neurons are tailored to encode slow viscoelastic and quasistatic events occurring in dermal and subdermal tissues (Westling and Johansson, 1987 [↗](#); Birznieks et al., 2001 [↗](#), 2009 [↗](#)). Furthermore, even in the absence of externally applied stimulation, they can exhibit ongoing impulsive activity, which suggests they are capable of monitoring inherent mechanical tension patterns in dermal and subdermal tissues in the skin in the unloaded state (Knibestöl, 1975 [↗](#); Johansson, 1978 [↗](#)). In particular, changes in the tension patterns via finger and hand movement without external stimulation can also modulate this activity. We believe that by constantly transmitting information related to tissues' viscoelastic state, SA-2 neurons could help keep the brain updated about the current mechanical state of body parts. In agreement with this view, peripheral nerve blocks affect the perceived image of body parts such as the arm and fingers (Inui et al., 2011 [↗](#); Walsh et al., 2015 [↗](#); Melzack and Bromage, 1973 [↗](#)). Likewise, although sensations elicited by electrical stimulation of single SA-2 neurons innervating the hand have been elusive (Kunesch et al., 1995 [↗](#); Ochoa and Torebjörk, 1983 [↗](#)), a recent study indicates that they can give rise to distinct sensations that include the experience of diffuse skin deformation (Watkins et al., 2022 [↗](#)). By continuously informing about viscoelastic state of the fingertips, SA-2 neurons could help the brain in computing accurate motor commands for interactions with objects in manipulation and haptic tasks by updating reference frames for interpreting information signaled by type I neurons.

## Consequences for haptic performance and perception

Although tactile information in general is crucial for planning and executing motor actions in such tasks, to our knowledge a possible influence of fingertip viscoelasticity on task performance has not been systematically investigated. Similarly, there is a lack of systematic investigation of potential effects of fingertip viscoelasticity on performance in tactile psychophysical tasks conducted during passive touch. Therefore, it is unclear whether viscoelasticity limits

performance or if it is compensated for in some way. Our findings indicate that the population of tactile neurons that innervate a fingertip encode continuous information about the fingertip's viscoelastic deformation state. This information could potentially aid the brain in managing the effects of viscoelasticity on tactile coding and fingertip actions. For instance, the brain could intermittently use this information to estimate the state of the fingertip during planning and evaluation of tactile-based actions (cf. Johansson and Flanagan, 2009 [↗](#)). It is also conceivable that the brain continuously represents the current deformation state of the fingertips using online population information. However, such processing might require considerable computational resources. In any case, the viscoelastic deformability of the fingertips plays a pivotal role in supporting the diverse functions of the fingers. For example, it allows for cushioned contact with objects featuring hard surfaces and allows the skin to conform to object shapes, enabling the extraction of tactile information about objects' 3D shapes and fine surface properties. Moreover, deformability is essential for the effective grasping and manipulation of objects. This is achieved, among other benefits, by expanding the contact surface, thereby reducing local pressure on the skin under stronger forces and enabling tactile signaling of friction conditions within the contact surface for control of grasp stability. Throughout, continuous acquisition of information about various aspects of the current state of the fingertip and its skin by tactile neurons is essential for the functional interaction between the brain and the fingers. In light of this, the viscoelastic memory effect on tactile signaling of fingertip forces can be perceived as a by-product of an overall optimization process within prevailing biological constraints.

## Materials and methods

### General procedure and study participants

Our study is based on data obtained from 21 females and 12 males (19-30 years of age) who participated after providing written informed consent in accordance with the Declaration of Helsinki, apart from pre-registration in a database. The Umeå University ethics committee approved the study. The general experimental methodology, procedure and apparatus have been described previously (Birznieks et al., 2001 [↗](#)), as well as other aspects of the same experimental data than those analyzed here (Birznieks et al., 2001 [↗](#); Jenmalm et al., 2003 [↗](#); Saal et al., 2009 [↗](#); Johansson and Birznieks, 2004 [↗](#)). Briefly, action potentials (spikes) in axons of single first-order tactile neurons that terminated in the distal segment of the index, middle or ring finger were recorded with tungsten needle electrodes inserted into the median nerve at the level of the upper arm 0.5-0.6 m from the fingertips (Vallbo and Hagbarth, 1968 [↗](#)). For neurons with cutaneous receptive fields on the distal segment of a finger, force stimuli were applied to its fingertip in five different directions by means of a custom-built robot. The fingertip was stabilized by gluing the nail to a firmly fixed metal plate. Force was transferred through a circular plane (30 mm diameter) that was centered on the midpoint of a line extending in the proximal-distal direction from the papillary whorl to the distal end of the finger. The stimulating surface was oriented parallel to the skin at this primary contact site (see **Figure 1B** [↗](#)), which was located approximately at the center of the flat part of the fingertip's volar surface and serves as a primary target for object contact in fine manipulation tasks engaging 'tip-to-tip' precision grips (Christel et al., 1998 [↗](#)). To ensure that friction between the contactor and the skin was sufficiently high to prevent slips, the surface was coated with silicon carbide grains (50–100 µm), approximating the finish of smooth sandpaper.

### Force stimuli

#### Force parameters

One of the five directions of force stimulation was normal (N) to the skin surface at the primary contact site and the other four were angled 20° to the normal direction in the radial (R), distal (D), ulnar (U), and proximal (P) directions, respectively. All force stimuli were superimposed on 0.2 N

background force normal to the skin and consisted of a force protraction phase (125 ms), a plateau phase at 4 N normal force (250 ms), and a force retraction phase (125 ms) (**Figure 1C**). In the four trials with a tangential force component, the tangential force was 1.4 N at the force plateau. The time course of the force changes followed a half-sinusoid (sine wave frequency of 4 Hz). The position and orientation of the stimulation surface in relation to the primary contact site was maintained when the loading contained tangential force components. That is, the friction between the stimulation surface and the skin was high enough to prevent frictional slips. The interval between successive fingertip loadings was 250 ms. Thus, the frequency of recurrent fingertip loadings (1.3 Hz) was representative of the frequency at which tactile events follow each other during dexterous object manipulation tasks (see e.g. [Draper, 1994](#); [Kunesch et al., 1989](#); [Teulings and Maarse, 1984](#)). Likewise, in trials with a tangential force component, the magnitudes, directions and time courses of the fingertip forces were similar to those employed when people use a precision grip to lift an object weighing 250–300 g ([Johansson and Westling, 1984](#); [Westling and Johansson, 1984](#)).

## Stimulation sequences

Two force stimulation sequences containing trials in each of the five different directions were delivered repeatedly. In the regular sequence, the trial order was the same (R, D, U, P, N) for all repetitions of the sequence. The regular sequence was presented 6 times, but to standardize the stimulation history for all trials only data obtained during the last five repetitions was included in the analysis (5 force directions  $\times$  5 trials). Immediately after completion of the regular sequence, the irregular sequence was delivered in which the trial order was systematically changed over repeats of the sequence. Each of the five loading directions (R, D, U, P and N) were presented five times in such a way that each loading was preceded once by loading in each of the five directions (5 force directions  $\times$  5 trials).

## Neural sample

The neurons recorded from were classified as fast-adapting type I (FA-1), fast-adapting type II (FA-2), slowly-adapting type I (SA-1), and slowly-adapting type II (SA-2) according to criteria described previously ([Johansson and Vallbo, 1983](#); [Vallbo and Johansson, 1984](#)). Briefly, FA afferents respond only to changes in skin deformation, whereas SA afferents show an ongoing response during periods of static skin deformation. Type I afferents (FA-1 and SA-1) possess small and well-delineated receptive fields if probed by light, pointed skin indentations, while the receptive fields of type II afferents (FA-2 and SA-2) are often large and poorly defined (see [Vallbo and Johansson, 1984](#), for further details).

The present analysis included 60 FA-1, 73 SA-1, and 41 SA-2 neurons terminating in the glabrous skin of the terminal segment of digits II, III or IV. Other functional aspects of these neurons have been explored in previous studies ([Birznieks et al., 2001](#); [Jenmalm et al., 2003](#); [Johansson and Birznieks, 2004](#); [Saal et al., 2009](#)). To ensure a reasonably balanced representation of the three neuron types in the sample, given the lower density of slowly adapting neuron types in fingertip compared with FA-1 neurons ([Johansson and Vallbo, 1979](#)), slowly adapting neuron types were intentionally prioritized during prerecording search for unitary action potentials. Initially, 10 FA-2 neurons were also included in the analysis. But their responsiveness during the experiment was remarkably low, and unlike the other neuron types, their responses were rarely affected by force stimuli. Specifically, only one of the observed FA-2 neurons responded during the force protraction phases. Due to the lack of clear stimulus-driven responses, FA-2 neurons were subsequently excluded from further analysis.

## Analysis

## Quantifying mechanical and neural response variability

We assessed the variability in displacements and velocities by calculating the standard deviation of the displacement and velocity signals at each time point over the two-dimensional tangential plane across repeated trials of the same force direction, using Euclidean distance as the metric. Variability was assessed separately for trials in the regular and the irregular sequence and averaged across all fingertips recorded from. Similarly, when assessing the variability of the neural responses, we calculated the standard deviation of the instantaneous firing rate across trials in the same force direction over time. Instantaneous firing rate was calculated as the inverse of the inter-spike interval at a given time point. Again, this analysis was run separately on data from the regular and the irregular sequence. The resulting standard deviation traces were first averaged over different force directions and then over all neurons from the same class.

## Predicting time-varying firing rates from contactor movements

This analysis was conducted in Python (v3.13) with *pandas* for data handling, *numpy* for numerical operations, and *scikit-learn* for model fitting and evaluation.

To assess how well individual neurons' time-varying firing rates could be predicted from simultaneous contactor movements, we fitted multiple linear regression models (see [Khamis et al., 2015](#), for a similar approach). This analysis focused on the force protraction phase of the irregular sequence, where neurons were most responsive and sensitive to stimulation history. Data from 100 ms before to 100 ms after the protraction phase (between -0.100 s and 0.225 s relative to protraction onset) were included for each trial. Neurons were included if they fired at least two action potentials during the force protraction phase and the following 100 ms in at least five of the 25 trials. This ensured sufficient variability in firing rates for meaningful regression analysis, resulting in 68 SA-1, 38 SA-2, and 51 FA-1 neurons being included.

Contractor position signals digitized at 400 Hz were linearly interpolated to 1000 Hz. Instantaneous firing rates, derived from action potentials sampled at 12.8 kHz, were resampled at 1000 Hz to align with position signals. A Gaussian filter ( $\sigma = 10$  ms, cutoff  $\sim 16$  Hz) was applied to the firing rate as well as to the position signals before differentiation. To account for axonal conduction (8–15 ms) and sensory transduction delays (1–5 ms), firing rates were advanced by 15 ms to align approximately with independent variables.

Regressions were performed using *scikit-learn*'s *Ridge* and *RidgeCV* regressors, which apply L2 regularization to mitigate overfitting. Hyperparameter tuning for the regularization parameter ( $\alpha$ ) was performed using *GridSearchCV* with a predefined range (0.001–1000.0), incorporating five-fold cross-validation to select the best value. To minimize overfitting risks, model performance was further validated with independent five-fold cross-validation (*KFold*), and  $R^2$  scores were computed using *cross\_val\_score*.

We constructed four linear regression models with increasing complexity: (1) Position-only, using three-dimensional contactor positions ( $P_x$ ,  $P_y$ ,  $P_z$ ); (2) Velocity-only, using three-dimensional velocities ( $dP_x$ ,  $dP_y$ ,  $dP_z$ ); (3) Combined, including all position and velocity signals (6 predictors); and (4) Interaction, including all signals and their two-way interactions (21 predictors). All features were standardized using *StandardScaler* to improve regularization and model convergence. *PolynomialFeatures* generated second-order interaction terms for the interaction model. Feature importance was evaluated with *permutation\_importance*, and simpler models were built using the most important features. These models were validated through cross-validation to assess retained explanatory power.

## Calculation of information transmission

To assess the amount of information about force direction conveyed in responses of individual neurons, for each neuron we calculated a lower bound on the information transmitted about both the current and the previous force direction. We used metric space analysis (Victor and Purpura, 1996), which employs a classifier on the neural response data to distinguish different force directions, calculates a confusion matrix from several runs of the classifier, and finally computes a lower bound on the mutual information from the confusion matrix. Details of the specific implementation used in the present study have been described previously (Saal et al., 2009). In short, we first sorted the trial data either by the force direction in the current trial (to assess coding of current force direction) or by the force direction in the previous trial (to assess whether neurons responded to the viscoelastic ‘memory’ of the previous trial). We then computed spike distances between all pairs of spike trains according to a spike-timing based distance metric that assesses the ‘cost’ for transforming the first spike train in the pair into the second, by adding or removing individual spikes, or shifting existing ones in time. For analyses of the protraction phase this spike distance was computed at a temporal resolution of 8 ms, as this was determined in a previous study to be close to the optimum for maximal information transmission for the present experimental data (Saal et al., 2009). For the analysis of SA-2 responses during the interstimulus period, we used a lower temporal resolution of 32 ms instead, to adjust for the lower firing rates of these neurons in the absence of externally applied stimulation. Each individual trial was then classified as originating from the force direction to which its average distance was lowest. In this way, a confusion matrix was generated for each neuron, with each entry denoting the number of times a neural response from a given force direction was classified as having originated from that direction or another one. Finally, we used the confusion matrix to compute a lower bound of the mutual information. To quantify the bias of this estimate, we reassigned neural responses from individual trials to random conditions and recalculated the mutual information. The bias term was set as the average of the outcome of 10 such random assignments and subtracted from the estimate of the mutual information. This analysis was run on successively longer time windows starting at stimulus onset (when calculating information during the protraction phase) or at the start of the interstimulus period (when calculating information for this period), respectively, and extending until the end of the considered time window in 5 ms increments. In this way, it could be assessed how neural coding of force direction evolved over time. The whole analysis was run twice on the irregular sequence data: once with stimuli grouped according to the current force direction (to assess the information conveyed about the stimulus), and a second time with stimuli grouped according to the preceding trial (to assess information conveyed in the neural responses about the past stimulus). The analysis was also run on the regular sequence; because effects due to previous and current stimulation cannot be distinguished in this data set due to the fixed trial order, this analysis yielded a single information value that represents the total amount of information available when past loading is held constant.

For the analysis covering the protraction phase, we also assigned each neuron to one of three groups, based on their information curves: those that primarily conveyed information about the current force direction (‘current’), those that conveyed information about the preceding force direction (‘previous’), and those that conveyed different types of information at different times during the protraction phase (‘mixed’). To do this, we first excluded neurons that responded too weakly or erratically, as assessed by whether they conveyed information (above 0) about either past or present force direction for at least 10 different time windows during the protraction phase. This left 53 FA-1, 67 SA-1, and 35 SA-2 neurons. For each time window containing non-zero information, we then measured whether more information was conveyed about the present or the past force direction. Neurons that conveyed information about the current force direction in 70% or more of time windows were classed as ‘current’; neurons with 30% or less were classed as ‘previous’; and all other neurons were classed as ‘mixed’.

## Analysis of SA-2 population responses

To derive a representation of the SA-2 population response, we took the spike train distances (see above) calculated at three different time points (in the middle and at the end of the interstimulus period at -0.125 s and 0 s relative to protraction onset, and at the end of the protraction phase at 0.125 s) for the irregular sequence and summed these distances across all 20 SA-2 neurons that were tonically active during the interstimulus period. This yielded a 25-by-25 matrix, with each entry denoting how dissimilar a given trial in the irregular sequence was from another one on the population level. We then employed multidimensional scaling, which aims to embed data points in a high-dimensional space such that their Euclidean distances adhere to those in the original distance matrix as closely as possible. We retained the first two dimensions of this space, resulting in each trial now occupying a position in a two-dimensional space. Finally, using Procrustes analysis we rotated and scaled this representation to match it to the two-dimensional skin deformations in the plane tangential to the fingertip surface at the three different time points. The main behind this analysis is that if time-varying skin deformations are encoded at the SA-2 population level, then more dissimilar skin deformations should lead to more dissimilar neural responses. The analysis tests the extent to which this idea is true.

## Statistics

As mechanical and neural measures were generally not normally distributed, we used non-parametric tests for all statistical analyses on these variables. Specifically, for all analyses comparing the regular and irregular sequences, we used paired Wilcoxon signed rank tests. When making multiple comparisons within the same analysis, we used Bonferroni corrections, in which case the resulting  $p$  values are reported as  $p_{corrected}$ .

To assess the effect of regression model and neuron type on  $R^2$  scores, a two-way mixed design ANOVA was conducted. The regression model (1–4) served as the within-group factor, and neuron type (FA-1, SA-1, SA-2) as the between-group factor. Correlation coefficients were Fisher transformed into Z-scores for parametric testing and averaging, then converted back to correlation coefficients for reporting. Results are expressed as coefficients of determination ( $R^2$ ). For post hoc comparisons, Tukey's HSD test was used to control the family-wise error rate.

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**Reviewer #1 (Public review):**

The authors investigate how the viscoelasticity of the fingertip skin can affect the firing of mechanoreceptive afferents and they find a clear effect of recent physical skin state (memory), which is different between afferents. The manuscript is extremely well-written and well-presented. It uses a large dataset of low threshold mechanoreceptive afferents in the fingertip, where it is particularly noteworthy that the SA-2s have been thoroughly analyzed and play an important role here. They point out in the introduction the importance of the non-linear dynamics of the event when an external stimulus contacts the skin, to the point at which this information is picked up by receptors. Although clearly correlated, these are different processes, and it has been very well-explained throughout. I have some comments and ideas that the authors could think about that could further improve their already very interesting paper. Overall, the authors have more than achieved their aims, where their results very much support the conclusions and provoke many further questions. This impact of the previous dynamics of skin affecting current state can be explored further in so many ways and may help us in understanding skin aging and the effects of anatomical changes of the skin better.

Comments on revised submission:

The authors have taken all my considerations into account and provided excellent responses to them. They have modified their paper accordingly, which improves its clarity even more. Very interesting work and I have no further comments.

<https://doi.org/10.7554/eLife.89616.2.sa2>

**Reviewer #2 (Public review):**

Summary:

The authors sought to identify the impact skin viscoelasticity has on neural signalling of contact forces that are representative of those experienced during normal tactile behaviour. The evidence presented in the analyses indicate there is a clear effect of viscoelasticity on the imposed skin movements from a force-controlled stimulus. Both skin mechanics and evoked afferent firing were affected based on prior stimulation, which has not previously been thoroughly explored. This study outlines that viscoelastic effects have an important impact on encoding in the tactile system, which should be considered in the design and interpretation of future studies. Viscoelasticity was shown to affect the mechanical skin deflections and stresses/strains imposed by previous and current interaction force, and also the resultant neuronal signalling. The result of this was an impaired coding of contact forces based upon previous stimulation. The authors may be able to strengthen their findings, by using the existing data to further explore the link between skin mechanics and neural signalling, giving a clearer picture than demonstrating shared variability. This is not a critical addition, but I believe would strengthen the work and make it more generally applicable.

#### Strengths:

- Elegant design of the study. Direct measurements have been made from the tactile sensory neurons to give detailed information on touch encoding. Experiments have been well designed and the forces/displacements have been thoroughly controlled and measured to give accurate measurements of global skin mechanics during a set of controlled mechanical stimuli.

- Analytical techniques used. Analysis of fundamental information coding and information representation in the sensory afferents reveals dynamic coding properties to develop putative models of the neural representation of force. This advanced analysis method has been applied to a large dataset to study neural encoding of force, the temporal dynamics of this, and the variability in this.

#### Weaknesses:

- Lack of exploration of the variation in neural responses. Although there is a viscoelastic effect which produces variability in the stimulus effects based on prior stimulation, it is a shame that the variability in neural firing and force induced skin displacements have been presented, and are similarly variable, but there has been no investigation of a link between the two. I believe with these data the authors can go beyond demonstrating shared variability. The force per se is clearly not faithfully represented in the neural signal, being masked by stimulation history, and it is of interest if the underlying resultant contact mechanics are.

#### Validity of conclusions:

The authors have succeeded in demonstrating skin viscoelasticity has an impact on skin contact mechanics with a given force and that this impacts on the resultant neural coding of force. Their study has been well designed and the results support their conclusions. The importance and scope of the work is adequately outlined for readers to interpret the results and significance.

#### Impact:

This study will have important implications for future studies performing tactile stimulation and evaluating tactile feedback during motor control tasks. In detailed studies of tactile function, it illustrates the necessity to measure skin contact dynamics to properly understand the effects of a force stimulus on the skin and mechanoreceptors.

<https://doi.org/10.7554/eLife.89616.2.sa1>

## Author response:

The following is the authors' response to the original reviews

### Public Reviews:

#### Reviewer #1 (Public Review):

*The authors investigate how the viscoelasticity of the fingertip skin can affect the firing of mechanoreceptive afferents and they find a clear effect of recent physical skin state (memory), which is different between afferents. The manuscript is extremely well-written and well-presented. It uses a large dataset of low threshold mechanoreceptive afferents in the fingertip, where it is particularly noteworthy that the SA-2s have been thoroughly analyzed and play an important role here. They point out in the introduction the importance of the non-linear dynamics of the event when an external stimulus contacts the skin, to the point at which this information is picked up by receptors. Although clearly correlated, these are different processes, and it has been very well-explained throughout. I have some comments and ideas that the authors could think about that could further improve their already very interesting paper. Overall, the authors have more than achieved their aims, where their results very much support the conclusions and provoke many further questions. This impact of the previous dynamics of the skin affecting the current state can be explored further in so many ways and may help us to better understand skin aging and the effects of anatomical changes of the skin.*

*At the beginning of the Results, it states that FA-2s were not considered as stimuli did not contain mechanical events with frequency components high enough to reliably excite them. Was this really the case, did the authors test any of the FA-2s from the larger dataset? If FA-2s were not at all activated, this is also relevant information for the brain to signal that it is not a relevant Pacinian stimulus (as they respond to everything). Further, afferent receptive fields that were more distant to the stimulus were included, which likely fired very little, like the FA-2s, so why not consider them even if their contribution was low?*

Thank you for bringing this up, we have now clarified in the text that while FA-2s did respond at a low rate during the experiment, their responses were not reliably driven by the force stimuli. In the Methods section we have included the following text:

“Initially, 10 FA-2 neurons were also included in the analysis. But their responsiveness during the experiment was remarkably low, and unlike the other neuron types, their responses were rarely affected by force stimuli. Specifically, only one of the observed FA-2 neurons responded during the force protraction phases. Due to the lack of clear stimulus-driven responses, FA-2 neurons were subsequently excluded from further analysis.”

*One question that I wondered throughout was whether you have looked at further past history in stimulation, i.e. not just the preceding stimulus, but 2 or 3 stimuli back? It would be interesting to know if there is any ongoing change that can be related back further. I do not think you would see anything as such here, but it would be interesting to test and/or explore in future work (e.g. especially with sticky, forceful, or sharp indentation touch). However, even here, it could be that certain directions gave more effects.*

This is a very interesting question! A discernible effect from the previous stimulus could persist at the end of the current stimulation (see Figure 4C), potentially influencing the next one—a 2-stimuli-back effect. Unfortunately, our experimental design did not allow for rigorous testing of this effect. While all possible pairs of stimulus directions were included in

immediately consecutive trials, this was not the case for pairs separated by additional trials. Hence, the combination of a likely weak effect and limited variation in history precluded a thorough analysis of a 2-stimuli-back effect. Future work should delve into the time course of the viscoelastic effect in greater detail.

*Did the authors analyze or take into account the difference between receptive field locations? For example, did afferents more on the sides have lower responses and a lesser effect of history?*

An investigation into the potential impact of the relationship between the receptive field location on the fingertip skin and the primary contact site of the stimulus surface revealed no discernible influence for SA-1 and SA-2 neurons. In contrast, FA-1 neurons, particularly those predominantly sensitive to the previous stimulation or displaying mixed sensitivity, exhibited a tendency to terminate near the primary stimulation site. We have added these observations to the text:

“We found no straightforward relationship between a neuron's sensitivity to current and previous stimulation and its termination site in fingertip skin. Specifically, there was no statistically significant effect of the distance between a neuron's receptive field center and the primary contact site of the stimulus surface on whether neurons signaled current, prior, or mixed information for SA-1 (Kruskal-Wallis test  $H(2)=3.86$ ,  $p=0.15$ ) or SA-2 neurons ( $H(2)=0.75$ ,  $p=0.69$ ). However, a significant difference emerged for FA-1 neurons ( $H(2)=8.66$ ,  $p=0.01$ ), indicating that neurons terminating closer to the stimulation site on the flat part of the fingertip were more likely to signal past or mixed information.”

*Was there anything different in the firing patterns between the spontaneous and non-spontaneously active SA-2s? For example, did the non-spontaneous show more dynamic responses?*

The firing patterns of both spontaneously and non-spontaneously active SA-2 neurons shared similarities in terms of adaptation and range of firing rate modulation in response to force stimuli, i.e., ‘dynamic response’. The distinction lay in the pattern of modulation of the firing rate associated with stimulus presentations. For spontaneously active SA-2 neurons, this modulation occurred around a significant background discharge, implying that a force stimulus could either decrease or increase the firing rate, depending on how it deformed the fingertip. This characteristic is well illustrated by the firing pattern of the neuron depicted in the lower panels of Figure 3D. Conversely, in non-spontaneously active SA-2 neurons, a force stimulus could only induce an increase in the firing rate or no change. Although the neuron depicted in the upper panels of Figure 3D exhibited some background activity, it serves to exemplify this characteristic. In the text, we have elucidated the dynamics of the SA-2 neuron response by highlighting that force stimulation can either decrease or increase the firing rate in neurons with spontaneous activity through the following addition/change:

“This increased variability was most evident during the force protraction phase where most neurons exhibited the most intense responses. Increased variability was also observed in instances where the dynamic response to force stimulation involved a decrease in the firing rate (lower panels of Figure 3D). This phenomenon was observed in SA-2 neurons that maintained an ongoing discharge during intertrial periods (cf. Fig. 2A). In these cases, the response to a force stimulus constituted a modulation of the firing rate around the background discharge, signifying that a force stimulus could either decrease or increase the firing rate depending on the prevailing stimulus direction.”

*Were the spontaneously active SA-2 afferents firing all the time or did they have periods of rest - and did this relate to recent stimulation? Were the spontaneously active SA-2s located in a certain part of the finger (e.g. nail) or were they randomly spread*

*throughout the fingertip? Any distribution differences could indicate a more complicated role in skin sensing.*

SA-2 neurons, in general, are well-known for undergoing significant post-stimulation depression (e.g., Knibestöl and Vallbo, 1970; Chambers et al., 1972; Burgess and Perl, 1973). In our force stimulations, this post-excitatory depression manifested as a reduced or absent response during the latter part of the stimulus retraction period for stimuli in directions that markedly excited the neuron. The excitability recovered when the fingertip relaxed during the subsequent intertrial period, and for "spontaneously active" neurons, the firing resumed (see examples in Figure 7A). Furthermore, some "spontaneously active" neurons could be silenced or exhibit a near-silent period during force stimulation for certain force directions, while the spontaneous firing returned during the upcoming intertrial period when the fingertip shape recovered (for example, see responses to stimulation in the proximal and especially ulnar directions in the top panel in Figure 7A).

Regarding the location of the receptive field centres of spontaneously active and non-spontaneously active SA-2 neurons on the fingertip we did not observe any obvious spatial segregation. To illustrate this, we have revised Figure 1A by color-marking SA-2 neurons that exhibited ongoing activity in intertrial periods, and the figure caption has been modified accordingly:

"Figure 1. Experimental setup. A. Receptive field center locations shown on a standardized fingertip for all first-order tactile neurons included in the study, categorized by neuron type. Purple symbols denote spontaneously active SA-2 neurons exhibiting ongoing activity without external stimulation."

*Did the authors look to see if the spontaneous firing in SA-2s between trials could predict the extent to which the type 1 afferents encode the proceeding stimulus? Basically, does the SA-2 state relate to how the type 1 units fire?*

We found no clear indications that the responses of FA-1 and SA-1 could be readily anticipated based on the firing patterns of SA-2 neurons.

*In the discussion, it is stated that "the viscoelastic memory of the preceding loading would have modulated the pattern of strain changes in the fingertip differently depending on where their receptor organs are situated in the fingertip". Can the authors expand on this or make any predictions about the size of the memory effect and the distance from the point of stimulation?*

We have explored this topic further in the text, referring to recent studies modeling essential aspects of fingertip mechanics. However, in our view, current models lack the capability to predict the specific nature sought by the reviewer. These models should include a detailed understanding of the intricate networks of collagen fibers anchoring the pulp tissue at the distal phalangeal bone and the nail. They should also consider potential inherent directional preferences of the receptor organs, attributed to their microanatomy. The text modifications are as follows:

"In addition to the receptor organ locations, the variation in sensitivity among neurons to fingertip deformations in response to both previous and current loadings would stem from the fingertip's geometry and its complex composite material properties. Possible inherent directional preferences of the receptor organs, attributed to their microanatomy, could also be significant. However, mechanical anisotropy, particularly within the viscoelastic subcutaneous tissue of the fingertip induced by intricately oriented collagen fiber strands forming fat columns in the pulp (Hauck et al., 2004), are likely to play a crucial role. This anisotropy would shape the dynamic pattern of strain changes at neurons' receptor sites,

intricately influencing a neuron's sensitivity not only to current but also to preceding loadings. Indeed, recent modeling efforts suggest that such mechanical anisotropy strongly influences the spatiotemporal distribution of stresses and strains across the fingertip (Duprez et al., 2024)."

Relatedly, we have included additional text to provide a more comprehensive explanation of the "bulk deformation" of the fingertip that occurs during the loadings:

"As pressure increases in the pulp, the pulp tissue bulges at the end and sides of the fingertip. Simultaneously, the tangential force component amplifies the bulging in the direction of the force while stretching the skin on the opposite side."

*In the discussion, it would be good if the authors could briefly comment more on the diversity of the mechanoreceptive afferent firing and why this may be useful to the system.*

The diversity in responses among neurons is instrumental in enhancing the information transmitted to the brain by averting redundancy in information acquisition. This diversity thereby contributes to an overall increase in information. We've included a brief statement, along with several references, underscoring this concept:

"The resulting diversity in the sensitivities of neurons might enhance the overall information collected and relayed to the brain by the neuronal population, facilitating the discrimination between tactile stimuli or mechanical states of the fingertip (see Rongala et al., 2024; Corniani et al., 2022; Tummala et al., 2023, for more extensive explorations of this idea)."

*Also, the authors could briefly discuss why this memory (or recency) effect occurs - is it useful, does it serve a purpose, or it is just a by-product of our skin structure? There are examples of memory in the other senses where comparisons could be drawn. Is it like stimulus adaptation effects in the other senses (e.g. aftereffects of visual motion)?*

We have expanded the concluding paragraph of the discussion, specifically delving into the question of whether the mechanical memory effect serves a deliberate purpose or is simply an incidental byproduct of our skin structure:

"In any case, the viscoelastic deformability of the fingertips plays a pivotal role in supporting the diverse functions of the fingers. For example, it allows for cushioned contact with objects featuring hard surfaces and allows the skin to conform to object shapes, enabling the extraction of tactile information about objects' 3D shapes and fine surface properties. Moreover, deformability is essential for the effective grasping and manipulation of objects. This is achieved, among other benefits, by expanding the contact surface, thereby reducing local pressure on the skin under stronger forces and enabling tactile signaling of friction conditions within the contact surface for control of grasp stability. Throughout, continuous acquisition of information about various aspects of the current state of the fingertip and its skin by tactile neurons is essential for the functional interaction between the brain and the fingers. In light of this, the viscoelastic memory effect on tactile signaling of fingertip forces can be perceived as a by-product of an overall optimization process within prevailing biological constraints."

*One point that would be nice to add to the discussion is the implications of the work for skin sensing. What would you predict for the time constant of relaxation of fingertip skin, how long could these skin memory effects last? Two main points to address here may be how the hydration of the skin and anatomical skin changes related to aging affect the results. If the skin is less viscoelastic, what would be the implications for the firing of mechanoreceptors?*

It is likely that the time constant depends to some extent on mechanical factors of the skin, which will likely change due to age or environmental factors. However, while these questions are intriguing, they fall outside the scope of the current study and we are not aware of studies that have addressed these issues directly in experiments either.

*How long does it take for the effect to end? Again, this will likely depend on the skin's viscoelasticity. However, could the authors use it in a psychophysical paradigm to predict whether participants would be more or less sensitive to future stimuli? In this way, it would be possible to test whether the direction modifies touch perception.*

Time constants for tissue viscoelasticity have been estimated to extend up to several seconds (see citations in the introduction). While direct perceptual effects could indeed be explored through psychophysical experimental paradigms, we are currently unaware of any studies specifically addressing the type of effect described in this study. In addition to the statement that, concerning manipulation and haptic tasks, "to our knowledge, a possible influence of fingertip viscoelasticity on task performance has not been systematically investigated," we have now also addressed tactile psychophysical tasks conducted during passive touch with the following sentence in the text:

"Similarly, there is a lack of systematic investigation of potential effects of fingertip viscoelasticity on performance in tactile psychophysical tasks conducted during passive touch."

#### **Reviewer #2 (Public Review):**

##### *Summary:*

*The authors sought to identify the impact skin viscoelasticity has on neural signalling of contact forces that are representative of those experienced during normal tactile behaviour. The evidence presented in the analyses indicates there is a clear effect of viscoelasticity on the imposed skin movements from a force-controlled stimulus. Both skin mechanics and evoked afferent firing were affected based on prior stimulation, which has not previously been thoroughly explored. This study outlines that viscoelastic effects have an important impact on encoding in the tactile system, which should be considered in the design and interpretation of future studies. Viscoelasticity was shown to affect the mechanical skin deflections and stresses/strains imposed by previous and current interaction force, and also the resultant neuronal signalling. The result of this was an impaired coding of contact forces based on previous stimulation. The authors may be able to strengthen their findings, by using the existing data to further explore the link between skin mechanics and neural signalling, giving a clearer picture than demonstrating shared variability. This is not a critical addition, but I believe would strengthen the work and make it more generally applicable.*

##### *Strengths:*

- Elegant design of the study. Direct measurements have been made from the tactile sensory neurons to give detailed information on touch encoding. Experiments have been well designed and the forces/displacements have been thoroughly controlled and measured to give accurate measurements of global skin mechanics during a set of controlled mechanical stimuli.*
- Analytical techniques used. Analysis of fundamental information coding and information representation in the sensory afferents reveals dynamic coding properties to develop putative models of the neural representation of force. This advanced analysis*

*method has been applied to a large dataset to study neural encoding of force, the temporal dynamics of this, and the variability in this.*

**Weaknesses:**

*- Lack of exploration of the variation in neural responses. Although there is a viscoelastic effect that produces variability in the stimulus effects based on prior stimulation, it is a shame that the variability in neural firing and force-induced skin displacements have been presented, and are similarly variable, but there has been no investigation of a link between the two. I believe with these data the authors can go beyond demonstrating shared variability. The force per se is clearly not faithfully represented in the neural signal, being masked by stimulation history, and it is of interest if the underlying resultant contact mechanics are.*

Thank you for this suggestion. We have added a new section investigating the link between skin deformation and neural firing in more depth via a simple neural model. Please see our answer below in the 'Recommendations' section for further details.

**Validity of conclusions:**

*The authors have succeeded in demonstrating skin viscoelasticity has an impact on skin contact mechanics with a given force and that this impacts the resultant neural coding of force. Their study has been well-designed and the results support their conclusions. The importance and scope of the work is adequately outlined for readers to interpret the results and significance.*

**Impact:**

*This study will have important implications for future studies performing tactile stimulation and evaluating tactile feedback during motor control tasks. In detailed studies of tactile function, it illustrates the necessity to measure skin contact dynamics to properly understand the effects of a force stimulus on the skin and mechanoreceptors.*

**Recommendations for the authors:**

**Reviewer #1 (Recommendations For The Authors):**

*(Very) minor comments*

*- The authors say at the beginning of the Results that, "The fourth type of tactile neurons in the human glabrous skin, fast adapting type II neurons...". Although generally written that there are four types of afferent in the glabrous skin, it would be better to state that these are low-threshold A-beta myelinated mechanoreceptive afferents, at least one time, as there are other types of afferent in the glabrous skin that respond to mechanical stimulation (e.g. low and high threshold C-fibers).*

This is now clarified at the start of the Results section:

"We recorded action potentials in the median nerve of individual low-threshold A-beta myelinated first-order human tactile neurons innervating the glabrous skin of the fingertip..."

*- Fig. 3: Could you add '(N)' as the measurement of force for Fig. 3A for Fz, Fy, and Fz? Also, please change 'Data was recorded' to 'Data were recorded' in the legend.*

Fixed.

- At the beginning of the Methods, you say that your study conforms to the Declaration of Helsinki, which actually requires pre-registration in a database. If you did not pre-register your study, please can you add '... in accordance with the Declaration of Helsinki, apart from pre-registration in a database'.

Thanks for making us aware of this. We have added the suggested qualifier to the ethics statement.

**Reviewer #2 (Recommendations For The Authors):**

*The neural representation/encoding of the actual displacement vectors would be a useful addition to the analyses. These vectors have been demonstrated to systematically change with the condition in the irregular series (Figure 2E) and will thus significantly act on the dynamics of induced mechanical changes in the skin with a given interaction force. Thus, it could be examined how the neurons code the magnitude of displacements as well as their direction. An evaluation of the extent to which the imposed displacement magnitudes are encoded in the neural responses would be a useful addition in explaining the signalling of the force events and how the central nervous system decodes these. Evaluating an alternative displacement encoding for comparison to pure force encoding may reveal more about how contact events are represented in the tactile system, which must decode these variable afferent signals to reconstruct a percept of the interaction. It could then be explored how the central nervous system may then scale the dynamic afferent responses based on the background viscoelastic state likely to be present in the SA-II afferent signals (Figure 7) for a context in which to evaluate the dynamic contact forces. This may of course be a complex relationship for the type-I afferents, where the underlying mechanical events evoking the firing (microslips not represented in global forces) have not been measured here. Such a model could be more widely applicable, as the skin viscoelasticity and displacement magnitudes are a straightforward measurement metric and could perhaps be used as a better proxy for neural signalling. This would allow the investigation of a wider variety of forces, and the study of the timing of the viscoelastic effect, both of which have been fixed here. This would give the work a broader impact, rather than just highlighting that this effect produces variability, it could reveal if this mechanical feature is structured in the neural representation. The categorical encoding/decoding tested here is specific to the stimuli used (magnitudes, intervals), but there is the possibility that this may be more generally applicable (within the bounds of forces/speeds) if the underlying basis of the variability in the signalling produced by the viscoelasticity is identified. Since the time course of the viscoelasticity has not been measured here (fixed forces and intervals), further study is required to fully understand the implications this has for a wider variety of situations.*

We agree that a better understanding of how the mechanical deformations are reflected in the resulting spike trains would be valuable. While ultimately a full understanding will need precise measurements of skin deformation across the whole fingertip to account for mechanical propagation to mechanoreceptor locations, relating the deformations at the contact location with neural firing patterns directly can provide useful hints into which aspects of deformation are encoded and how. To this end, we ran a new analysis that aimed to predict the time-varying neural responses directly from the recorded mechanical movements of the contactor.

Below we have reproduced the new results and methods text along with the additional figures for this analysis. Note that we have also added text in the Discussion to interpret these findings in the context of our other results.

New section in Results titled Predicting neural responses from contactor movements: “The similarity in the history-dependent variation in neural firing and fingertip deformation at a given force stimulus suggests that neuronal firing is determined by how the fingertip deforms rather than the applied force itself. However, this similarity does not clarify the relationship between fingertip deformation dynamics and neural signaling. To investigate further, we fit cross-validated multiple linear regression models to evaluate how well distinct aspects of contactor movement could predict the time-varying firing rates of individual neurons during the protraction phases of the irregular sequence. The models used predictors based on (1) the three-dimensional position of the contactor, (2) its three-dimensional velocity, (3) a combination of position and velocity signals, and, finally, (4) position and velocity signals along with all possible two-way interactions between them, capturing potentially complex relationship between fingertip deformations and neural signaling.

Comparing the variance explained ( $R^2$ ) by each regression model for each neuron type revealed clear differences between the models (Figure 5A). A two-way mixed design ANOVA, with regression model as within-group effects and neuron type as a between-group effect revealed a main effect of model on variance explained ( $F(3,462) = 815.5$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.84$ ). Model prediction accuracy overall increased with the number of predictors, with the two-way interaction model outperforming all others ( $p < 0.001$  for all comparisons, Tukey’s HSD). Additionally, a significant main effect of neuron type ( $F(2,154) = 29.8$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.28$ ) and a significant interaction between regression model and neuron type were observed ( $F(6,462) = 50.8$ ,  $p < 0.001$ ,  $\eta_p^2 = 0.40$ ).

For neuron type, model predictions were most accurate for SA-2 neurons, followed by SA-1 neurons, with FA-1 neurons showing the lowest accuracy ( $p < 0.003$  for all comparisons, Tukey’s HSD). The interaction between model and neuron type revealed distinct patterns. For SA-1 and SA-2 neurons, position-only and velocity-only models had similar prediction accuracy ( $p \geq 0.996$ , Tukey’s HSD) with no significant differences between these neuron types ( $p \geq 0.552$ , Tukey’s HSD). FA-1 neurons performed poorly with the position-only model but showed higher accuracy with the velocity-only model ( $p < 0.001$ , Tukey’s HSD) and better than SA-1 neurons ( $p = 0.006$ , Tukey’s HSD). Models combining position and velocity predictors (without interactions) surpassed both position-only and velocity-only models for SA-1 and SA-2 neurons ( $p < 0.001$ , Tukey’s HSD). Overall, the differences between neuron types broadly match their tuning to static and dynamic stimulus properties.

The two-way interaction model, accounting for most variance in neural responses, produced mean  $R^2$  values of 0.75 for FA-1, 0.88 for SA-1, and 0.91 for SA-2 neurons (Figure 5A). To evaluate the contribution of the different predictors, we ranked them using the permutation feature importance method, focusing on the six most important ones. Regression analyses using only these variables explained almost all of the variance explained by the full model, with a median  $R^2$  reduction of just 0.055 across all neurons. Across all neuron types, at least half included all three velocity components (dPx, dPy, dPz) among the top six, with FA-1 neurons showing the highest prevalence (Figure 5B). Interactions between normal position (Pz) and each velocity component were also frequently observed, while interactions involving tangential position and velocity components were less common. Interactions among velocity components were relatively well represented, followed by interactions limited to position components. Position signals were generally less represented, except for normal position (Pz) in slowly adapting neurons, where it appeared in 50% of SA-1 and 68% of SA-2 neurons. Despite these broad trends, important predictors varied widely across ranks even within a given neuron class (see Figure 5-figure supplement 1), and even the most frequent variables appeared in only a subset of cases, suggesting broad variability in sensitivity across neurons.”

New methods paragraph titled Predicting time-varying firing rates from skin deformations:

“This analysis was conducted in Python (v3.13) with pandas for data handling, numpy for numerical operations, and scikit-learn for model fitting and evaluation.

To assess how well individual neurons' time-varying firing rates could be predicted from simultaneous contactor movements, we fitted multiple linear regression models (see Khamis et al., 2015, for a similar approach}. This analysis focused on the force protraction phase of the irregular sequence, where neurons were most responsive and sensitive to stimulation history. Data from 100 ms before to 100 ms after the protraction phase (between -0.100 s and 0.225 s relative to protraction onset) were included for each trial. Neurons were included if they fired at least two action potentials during the force protraction phase and the following 100 ms in at least five of the 25 trials. This ensured sufficient variability in firing rates for meaningful regression analysis, resulting in 68 SA-1, 38 SA-2, and 51 FA-1 neurons being included.

Contractor position signals digitized at 400 Hz were linearly interpolated to 1000 Hz. Instantaneous firing rates, derived from action potentials sampled at 12.8 kHz, were resampled at 1000 Hz to align with position signals. A Gaussian filter ( $\sigma = 10$  ms, cutoff  $\sim 16$  Hz) was applied to the firing rate as well as to the position signals before differentiation. To account for axonal conduction (8–15 ms) and sensory transduction delays (1–5 ms), firing rates were advanced by 15 ms to align approximately with independent variables.

Regressions were performed using scikit-learn's Ridge and RidgeCV regressors, which apply L2 regularization to mitigate overfitting. Hyperparameter tuning for the regularization parameter (alpha) was performed using GridSearchCV with a predefined range (0.001–1000.0), incorporating five-fold cross-validation to select the best value. To minimize overfitting risks, model performance was further validated with independent five-fold cross-validation (KFold), and  $R^2$  scores were computed using `cross_val_score`.

We constructed four linear regression models with increasing complexity: (1) Position-only, using three-dimensional contactor positions ( $P_x$ ,  $P_y$ ,  $P_z$ ); (2) Velocity-only, using three-dimensional velocities ( $dP_x$ ,  $dP_y$ ,  $dP_z$ ); (3) Combined, including all position and velocity signals (6 predictors); and (4) Interaction, including all signals and their two-way interactions (21 predictors). All features were standardized using StandardScaler to improve regularization and model convergence. PolynomialFeatures generated second-order interaction terms for the interaction model. Feature importance was evaluated with `permutation_importance`, and simpler models were built using the most important features. These models were validated through cross-validation to assess retained explanatory power.”

*Minor:*

*- It would be useful to add a brief description of the material aspects of the contactor tip to the methods (as per Birznieks 2001).*

We have added the following statement:

“To ensure that friction between the contactor and the skin was sufficiently high to prevent slips, the surface was coated with silicon carbide grains (50–100  $\mu\text{m}$ ), approximating the finish of smooth sandpaper.”

*- The axes labelling on Figure 3A and legend description is ambiguous, probably placing the  $P_x$ ,  $P_y$ , and  $P_z$  labels on the far left axes and the  $F_x$ ,  $F_y$ , and  $F_z$  on the right side of the far right axes would make this clearer.*

Label placement has been improved along with some other minor fixes.

- For the quasi-static phase analysis, the phrase "absence of loading" used in reference to the interstimulus period and SA-II afferents does not seem to be a correct description. The finger is still loaded (at least in the normal direction), with a magnitude of imposed displacement that counteracts the viscoelastic force exerted by the skin mechanics of the fingertip. Although there is a zero net-force load, a mechanical stimulus is still being actively applied to the skin.

We have changed the wording throughout the text and now consistently refer either to the "interstimulus period" directly or to an "absence of externally applied stimulation" to avoid confusion.

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